MIL-S-43910

21 November 1974

MILITARY SPECIFICATION

SPOT REMOVER: (FOR REMOVAL OF OIL AND GREASE FROM CLOTHING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a liquid spot remover for removing oil and grease stains from washable fabrics, including durable press, when applied to stains prior to laundering.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-26701	-	Bottles Screw Cap, and Carboys, Polyethylene Plastic

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6850

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LAWS AND REGULATIONS

16 CFR 1500 - Federal Hazardous Substances Act

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

US Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards

ASTM D56 - Testing for Flash Point by Tag Closed Tester

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Freight Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. The materials used for the spot remover shall be a clear, single-phase liquid consisting of a solution of an oil-soluble nonionic surfactant in an aliphatic petroleum distillate and shall conform to the requirements in table I.

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TABLE I. Spot remover (chemical and physical properties)

Characteristics	Percent by weight Minimum	Maximum
Matter nonvolatile at 95°C	4.9	5.4
Aliphatic petroleum distillate 1/	94.5	95.0
Oil-soluble nonionic surfactant 2/	5.0	5.5
Flash point, tag closed cup, °F	138°	- -
Chlorinated solvents, qualitative test	"negative" test	

1/ Shall conform to the requirements of P-D-680 (type II).

2/ See 3.2.

3.2 Nonionic surfactant. The surfactant contained in the spot remover shall be an oil-soluble surface active agent prepared by the ethoxylation of petroleum derived linear alcohols in the C_{11} to C_{15} range and containing an average of 5 moles of ethylene oxide. The surfactant shall meet the requirements in table II.

TABLE II. Nonionic surfactant (chemical and physical properties)

Characteristics	Requirement Minimum	Maximum
Molecular weight	400	425
Water, percent	- -	0.5
Specific gravity, 25°/25°C	0.960	0.980
pH, 1 percent solution at 25°C*	6	8
Biodegradability, percent	90	- -

* Determined in a 10:6 isopropanel: water solution.

3.3 Odor. The spot remover shall have a mild odor essentially that of Stoddard Solvent.

3.4 Labeling. In addition to the markings specified in 5.3, each container shall be durably and legibly labeled with the following directions for use, information, precautions, and conforming to the Federal Hazardous Substances Act:

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Spot Remover, Laundry
(For Removal of Oil and Grease Stains)

DIRECTIONS FOR USE

1. Apply the spot remover to oil and grease stains using a paint brush or other soft bristled brush.

2. Thoroughly wet out the stained fabric.

3. Allow to set for about 5 minutes, then place the treated garments in washer and launder in accordance with the established washing formula.

CAUTION

CONTAINS FLAMMABLE SOLVENT
KEEP AWAY FROM FLAME AND EXTREME HEAT
AVOID PROLONGED SKIN CONTACT
USE ONLY WITH ADEQUATE VENTILATION
AVOID PROLONGED BREATHING OF VAPOR
KEEP OUT OF REACH OF CHILDREN
CONTAINS PETROLEUM DISTILLATE
IF SWALLOWED, DO NOT INDUCE VOMITING;
CALL PHYSICIAN IMMEDIATELY

3.5 Workmanship. The spot remover shall be a clear liquid, free from sediment, suspended matter and undissolved water as determined by visual inspection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs and at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. A lot shall consist of all the spot remover manufactured in one batch. The lot shall be expressed in units of filled unit containers of the same capacity for examination in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination for preparation for delivery under 4.3.1.3.

4.3.1.1 Examination of contents of the unit container. The sample unit for this examination shall be the contents of one filled unit container, as applicable.

Examine	Defect
Odor	Rancid, fishy, or acrid
Appearance	Presence of sediment Not clean Not uniform
Container	Any damage which may result in a leak
Special marking (including instructions)	Missing, incorrect, illegible or incomplete

4.3.1.2 Examination of the end item for net contents. The sample unit for this examination shall be one filled container, as applicable. The lot shall be unacceptable if the average net contents per container is less than specified or indicated for all sample units examined.

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Examination of preparation for delivery requirements. An examination
le to determine that the packaging, packing, and marking complies
:tion 5 requirements. Defects shall be scored in accordance with the
The sample unit shall be one shipping container fully prepared for
rcept that it need not be closed. Defects of closure listed below
mined on shipping containers fully prepared for delivery. The lot
be the number of shipping containers in the end item inspection lot.

	Defect
terior and	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
	Any component missing, damaged or not as specified.
	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling.
	Bulged or distorted containers.
	Number of bottles of spot remover per container is more or less than required

Inspection levels and AQLs for examination. The inspection levels
ining the sample size, and the AQLs, expressed in defects per one
its, shall be as follows:

n paragraph	Inspection level	AQL
1.1	I	2.5
1.2	S-3	NA
1.3	S-2	2.5

esting of the end item. The spot remover shall be tested for the
stics specified in table III. The lot size shall be expressed in
cleaning compound. The sample unit shall be one quart of spot remover
it random throughout the lot. The lot shall be unacceptable if any
.t fails to meet any test requirement specified.

(quarts)	Sample size (quarts)
is	2
and including 22,000	3
more	5

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TABLE III

INSTRUCTIONS FOR TESTING OF THE END ITEM

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspection Level	AQL
	Requirement	Test Method	Individual Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Matter nonvolatile at 95°C	Table I	4.4.1	X		2		0.1		
Aliphatic petroleum distillate	Table I	1/	X						
Oil-soluble nonionic surfactant	Table I	1/	X						
Flash point, tag closed cup	Table I	2/	X			X			
Chlorinated solvent	Table I	4.4.2	X		2	X			

1/ The supplier shall furnish a certificate of compliance covering this requirement.

2/ ASTM Method D56

AMXRE Form 598
8 February 1965

MIL-S-43910

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4.4 Test procedures.

4.4.1 Matter nonvolatile at 95°C. Weight to the nearest 0.1 milligram a 10 ml sample of the spot remover into a tared, low-form weighing bottle (70 mm inside diameter and 33 mm height). Remove the glass stopper and place sample on a billowing steam bath for one hour. Remove from steam bath and place in a forced air circulating oven at 95 ± 1°C for 15 minutes. Remove sample from oven, replace glass stopper, cool in desiccator and weigh the nonvolatile matter. Calculate as follows:

$$\text{Percent matter nonvolatile at } 95^{\circ}\text{C} = \frac{\text{Weight of nonvolatile matter} \times 100}{\text{Weight of sample}}$$

4.4.2 Detection of presence of chlorinated solvents. The following qualitative test shall be applied to the spot remover:

a. Adjust a Bunsen burner to give a colorless flame. Heat a piece of copper wire in the flame and plunge it into ethyl alcohol (two inches in a small test tube is sufficient). When the wire itself imparts no color to the flame, it is clean.

b. Dip the cool, clean wire into the spot remover and insert it into the edge of the flame. Observe closely at the moment of insertion.

c. A green color imparted to the flame, sometimes only a momentary flash, is due to the volatilization of copper chloride and is considered a "positive" test for chlorinated solvents.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The spot remover shall be packaged in 1 pint or 1 gallon capacity polyethylene bottles, as specified (see 6.2). Each polyethylene bottle shall be fabricated and closed in accordance with the applicable requirements of MIL-B-26701, except the 1 pint capacity bottles shall be fitted with a sprayer closure unit (see 6.3). Twelve filled 1 pint capacity bottles shall be further packaged in a snug-fitting fiberboard box conforming to style RSC, type CF (variety SW), class domestic, grade 200 of PPP-B-636. Each box shall be securely closed in accordance with the appendix of the box specification. Filled 1 gallon capacity bottles do not require additional packaging.

5.1.2 Level C (commercial packaging). The spot remover shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Forty-eight 1 pint or six 1 gallon capacity bottles of the spot remover, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC or RSC-L (as applicable), grade V2s of PPP-B-636. The inside of each container packed with 1 gallon bottles of spot remover shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. In addition, each container fitted with a box liner shall be provided with top and bottom pads fabricated of weather resistant grade fiberboard material. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B. Forty-eight 1 pint or six 1 gallon capacity bottles of spot remover, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC or RSC-L (as applicable), type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container packed with 1 gallon bottles of spot remover shall be fitted with a box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. In addition, each container fitted with a box liner shall be provided with top and bottom pads fabricated of domestic grade fiberboard material. The pads shall measure 1/4 inch less than the inside length and width of the box liner. Each fiberboard container shall be closed in accordance with method II as specified n the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C (commercial packing). The spot remover, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking. In addition to the labeling specified in 3.4 and any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The spot remover covered by this specification is intended for use by institutional and home laundries for the removal of oil and grease stains from washable fabric items including durable press garments, sleeping bags, duffle bags and light weight loadcarrying equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(b) Whether spot remover shall be packaged in 1 pint or 1 gallon capacity polyethylene bottles (see 5.1.1).
(c) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 The sprayer closure unit specified in 5.1.1 for the 1 pint capacity bottles shall conform to S. H. ANSELL MODEL NO. 5 SPRAYER, or equal.

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User activities:

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Preparing activity:

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-43910-SPOT REMOVER: (FOR REMOVAL OF OIL AND GREASE FROM CLOTHING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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